FINAL YEAR PROJECT / SEMINAR

Multi-crop disease detection with federated averaging

BANANA · CAULIFLOWER · GROUNDNUT · RADISH

AGRI VISION

Federated multi-crop disease detection using one-stage object detectors

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ABSTRACT

Agri Vision studies crop disease and pest-symptom detection using bounding boxes and class labels. The dataset covers banana, cauliflower, groundnut and radish across 24 classes. Four one-stage detectors—YOLOv8n, SSDLite, SSD–MobileNetV3 and EfficientDet-D0—are trained through separate three-client federated-averaging simulations and assessed with a shared pooled evaluator.

PROBLEM STATEMENT

- Small lesions and similar symptoms make visual diagnosis difficult.
- Uneven class representation and variable annotation quality affect detection.
- A useful detector must balance disease recognition, localisation and computing cost.

AIM AND OBJECTIVES

Assess lightweight one-stage detectors for federated crop-symptom detection.

- Use fixed 24-class train, validation and held-out test splits.
- Compare four complete detectors through independent experiments.
- Report precision, recall and AP; measure inference speed and model size.

METHODOLOGY

One shared dataset → fixed three-client partitions → separate detector experiments

CLIENT 1 • LOCAL TRAINING | **CLIENT 2 • LOCAL TRAINING** | **CLIENT 3 • LOCAL TRAINING**

TRAINING-IMAGE-WEIGHTED FEDAVG → GLOBAL DETECTOR

YOLOv8n | **SSDLite** | **SSD–MobileNetV3** | **EfficientDet-D0**

Local training → aggregate → pooled validation → save checkpoints → repeat

Three logical clients simulated sequentially. Current runs use custom FedAvg; Flower integration is planned.

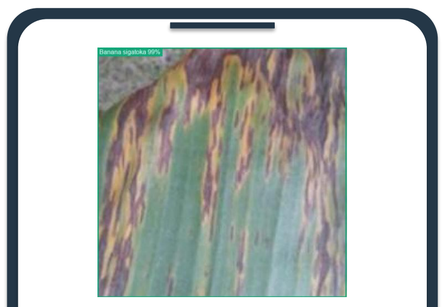
RESULTS

DETECTION PERFORMANCE • VALIDATION

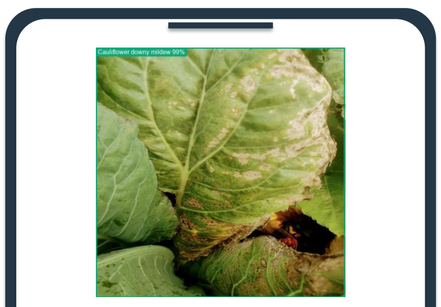
Detector	Precision	Recall	mAP50	mAP50–95
YOLOv8n	60.41%	55.99%	69.47%	45.34%
SSDLite	68.28%	55.84%	65.88%	45.20%
SSD-MobileNetV3	55.52%	55.01%	57.57%	35.20%
EfficientDet-D0	66.05%	78.02%	84.99%	59.56%

Each detector is evaluated on the same 24-class validation split. The comparison considers correct disease recognition, missed symptoms and bounding-box localisation.

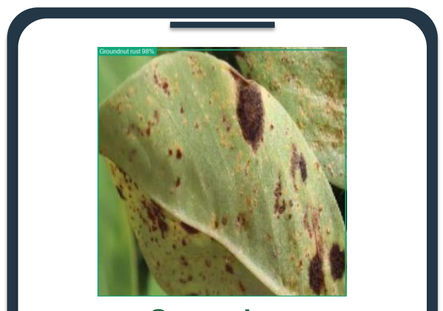
Same 4,157-image validation split. P/R: confidence 0.25, IoU 0.50. Selected model performance is reported using detection metrics. Held-out test evaluation and deployment profiling are planned.



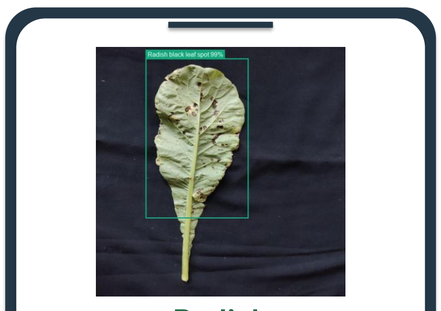
Banana
YOLOv8n



Cauliflower
YOLOv8n



Groundnut
YOLOv8n



Radish
YOLOv8n

Selected model predictions • disease labels and confidence

TECHNOLOGIES USED

SUMMARY

Agri Vision combines multi-crop disease localisation with three-client federated averaging. Four one-stage detectors are compared on a shared dataset. The linked picture demonstration presents actual predictions across all four crops.

SCAN TO VIEW
DETECTION EXAMPLES

DATA: 14,514 train / 4,157 validation / 2,091 test • 24 classes across four crops

Three simulated clients • Four one-stage detectors • Multi-crop disease localisation